

Healthcare IAM Project – Network Design

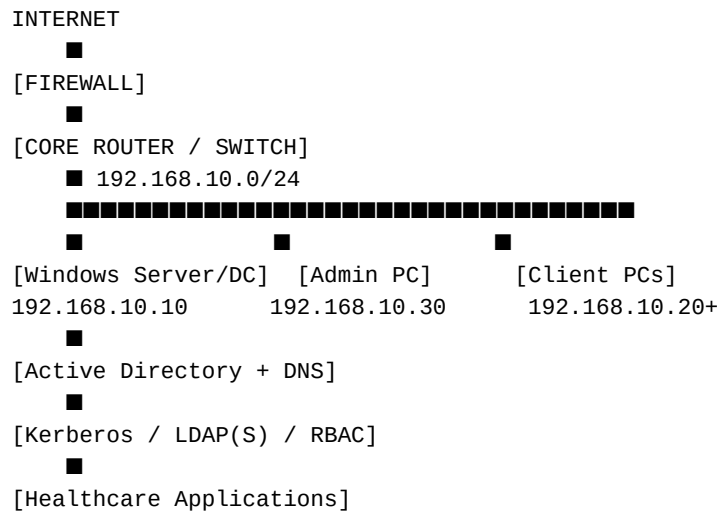
Design Network Protocols and Architectures – Rubric Response

1. Project Network Scenario

The healthcare IAM lab will use a small secure hospital network supporting a Windows Server/Active Directory environment, administrator workstation, and client systems. The network design connects directly to the project's IAM, RBAC, security-group, and access-control activities.

Network: 192.168.10.0/24 **Gateway:** 192.168.10.1 **Domain Controller:** 192.168.10.10

2. Network Diagram



3. OSI Model – One Device

Layer	OSI Layer	Protocol / Technology	IAM / Healthcare Use
7	Application	DNS, LDAP, Kerberos, HTTP	Directory, authentication and application communication
6	Presentation	TLS/SSL, encryption	Protects authentication and session data
5	Session	Kerberos session management	Maintains authenticated sessions
4	Transport	TCP / UDP	End-to-end communication
3	Network	IPv4 / ICMP	IP addressing, routing and connectivity testing
2	Data Link	Ethernet	Local LAN communication
1	Physical	Ethernet / Wi-Fi	Physical network connection

4. TCP/IP Model

TCP/IP Layer	Protocols	Purpose
Application	DNS, LDAP/LDAPS, Kerberos, HTTPS	IAM, authentication, directory and application communication
Transport	TCP / UDP	End-to-end communication
Internet	IPv4, ICMP	Addressing, routing and connectivity
Network Access	Ethernet	Local network communication

OSI → TCP/IP Mapping

OSI Layers 7–5 → TCP/IP Application | OSI Layer 4 → TCP/IP Transport | OSI Layer 3 → TCP/IP Internet | OSI
Layers 2–1 → TCP/IP Network Access

5. Subnetting – One Subnet

Item	Value
Network Address	192.168.10.0
Subnet Mask	255.255.255.0
CIDR	/24
First Usable IP	192.168.10.1
Last Usable IP	192.168.10.254
Broadcast Address	192.168.10.255
Usable Hosts	254

Example IP Assignments

Device	IP Address	Purpose
Router / Gateway	192.168.10.1	Default gateway
Windows Server / Domain Controller	192.168.10.10	Active Directory + DNS
Client PC	192.168.10.20	Domain client
Admin PC	192.168.10.30	IAM administration

6. Secure Network Architecture

The Internet connects to a firewall before reaching the internal network. The firewall controls inbound and outbound traffic. The internal /24 network contains the Windows Server/Domain Controller, administrator workstation, and client computers. Active Directory provides centralized identity management, while RBAC and least privilege determine what each healthcare role can access.

7. Specific Security Protocols

Security Control / Protocol	Purpose in the Healthcare IAM Lab
Kerberos	Secure authentication for Active Directory users.
LDAP / LDAPS	Directory communication; LDAPS adds encrypted communication.
HTTPS + TLS	Encrypts communication between users and healthcare applications.
IPsec / SSL VPN	Provides encrypted remote access for authorized users.
Firewall	Controls and filters traffic between the Internet and internal network.
RBAC	Provides access based on healthcare job role.
Least Privilege	Limits users to only the access required for their responsibilities.

8. Rubric-Ready Answer

Design Network Protocols and Architectures: The healthcare IAM lab will use a secure 192.168.10.0/24 internal network containing a Windows Server Domain Controller, administrator workstation, and client systems. The network will be protected by a firewall and use secure protocols including Kerberos for Active Directory authentication, LDAP/LDAPS for directory services, HTTPS/TLS for encrypted application communication, and VPN encryption for authorized remote access. The network diagram maps the Windows Server communication stack to both the seven-layer OSI model and four-layer TCP/IP model. RBAC and least-privilege principles will control which healthcare users can access systems and resources.

Project connection: This network design supports the project's healthcare IAM environment, including Active Directory, security groups, RBAC, authentication, and access-control activities.